Novelty P3 Amplitude Differentiation for Food and Non-food Stimuli

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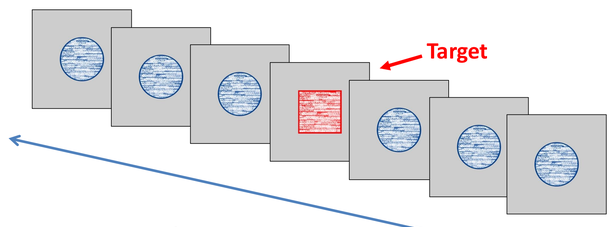
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INTRODUCTION

- In recent years there has been a dramatic increase in the prevalence of overweight and obese individuals around the world. In our obesogenic environment the abundance of food cues may result in unhealthy food-related decision making. While the etiology of obesity is multifactorial, we postulate that increased *attention* to food cues may contribute to over-eating.
- The present study examined novelty P3 responses to biologically significant stimuli (food) and compared them other neutral stimuli. Modulation of novelty P3 responses to infrequently occurring stimuli has been suggested to reflect variation in orienting and attention-capture of those rare events[1]. Specifically, novelty P3 amplitude reflects stimulus saliency and the degree in which there will be further processing [2],[3].
- In this study, young adults performed a visual novelty oddball task. They were asked to press a button for rare (16%) target stimuli presented among frequent stimuli. They were not informed about the occurrence of infrequent (16%) novel stimuli (food and neutral objects). Half of the food stimuli were low calorie food images and half were high calorie food images.
- We expect that if food, and especially high calorie food, is biologically significant to the subjects, it will elicit larger novelty P3 amplitude than non-food cues.

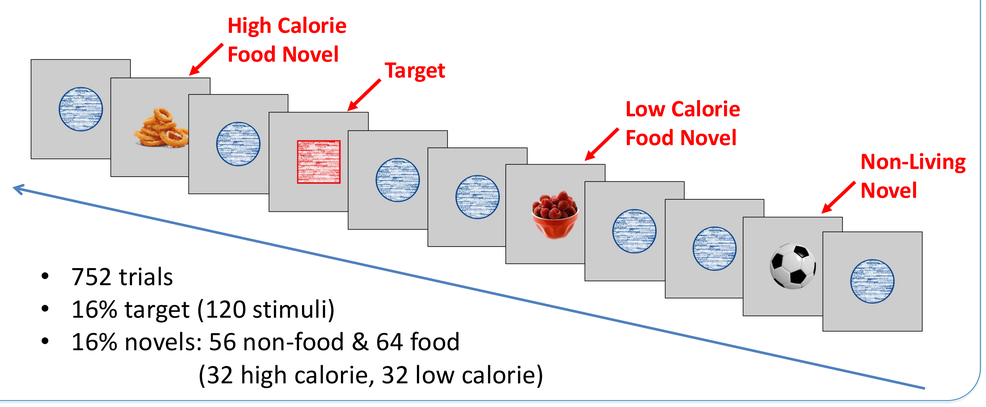
VISUAL ODDBALL TASK

Standard Oddball



- 150 trials
- 16% target (24 stimuli)

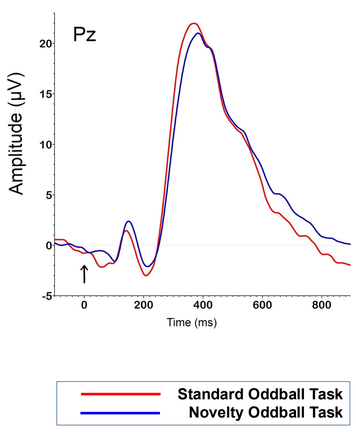
Novelty Oddball



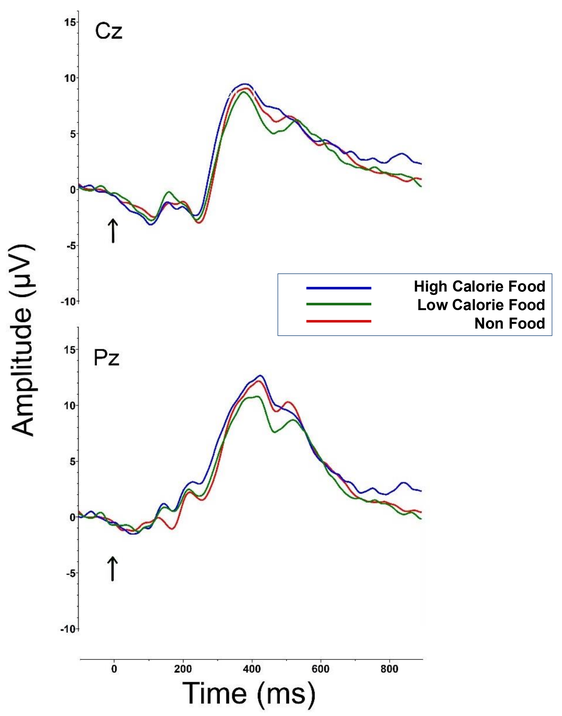
- 752 trials
- 16% target (120 stimuli)
- 16% novels: 56 non-food & 64 food
  (32 high calorie, 32 low calorie)

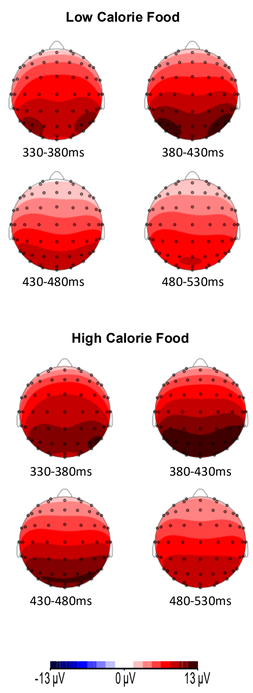
RESULTS

TARGET ERP



NOVEL ERP





- Both food and non-food images elicited novelty P3 responses which did not differ significantly in their magnitudes.
- High calorie food images elicited significantly larger novelty P3 than low calorie food images in the 380-480 ms time interval. It was particularly pronounced in the parietal and posterior scalp sites.

REFERENCES

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METHODS

Participants:
- 13 healthy lean (BMI=21.73 kg/m^2) young-adults (mean age = 25.31 years, 5 females) participated in the study. Subjects admitted to study if they reported no dietary restriction or excessive exercise regimen.

Stimuli:
- Pictures were selected from both the Food-Pictures database [4] and from the internet.
- Stimuli were presented on a screen at a distance of 1 m from the subject.

EEG recordings and preprocessing:
- EEG at 64 scalp sites (10-20 system) were recorded using ActiCAP (Brain Product) with nose as reference with 500 Hz sampling rate. Bipolar VEOG and HEOG were recorded to monitor eye movements.
- Data processing included filtering, artifact removal by visual inspection, and ICA was used for ocular correction (Vision Analyzer).

ERP analysis:
- EEG data were segmented to 1000 s epochs with 100 ms baseline correction pre-stimulus. ERP were generated by averaging epochs for each stimulus type. Voltage averages at 50 ms bin intervals between 330 – 530 ms (novelty P3 peak) were computed for each average type in 4 midline electrode sites (Fz, Cz, Pz, Oz). Repeated Measure ANOVA was used to compare between stimulus conditions.

CONCLUSIONS

- Novelty P3 amplitudes were similar for both food and non-food images, rejecting our initial hypothesis that biologically significant images will elicit larger novelty P3 amplitude.
- The difference in novelty P3 to high and low calorie food images support our hypothesis that high calorie food is biologically more salient or more important from an evolutionary perspective.
- The fact that we see difference in novelty P3 magnitude based on calorie content but not based on image category (non-food vs. food) is difficult to explain but may be due to the small sample size.
- The simplicity of this paradigm can be used in patients with eating disorders to assess the role of early attentional processes when exposed to high and low calorie food cues.

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